

----- PROJECT 1 -----

Airbnb Rio de Janeiro:
Where Should You Stay?

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Data Science & Machine Learning Bootcamp

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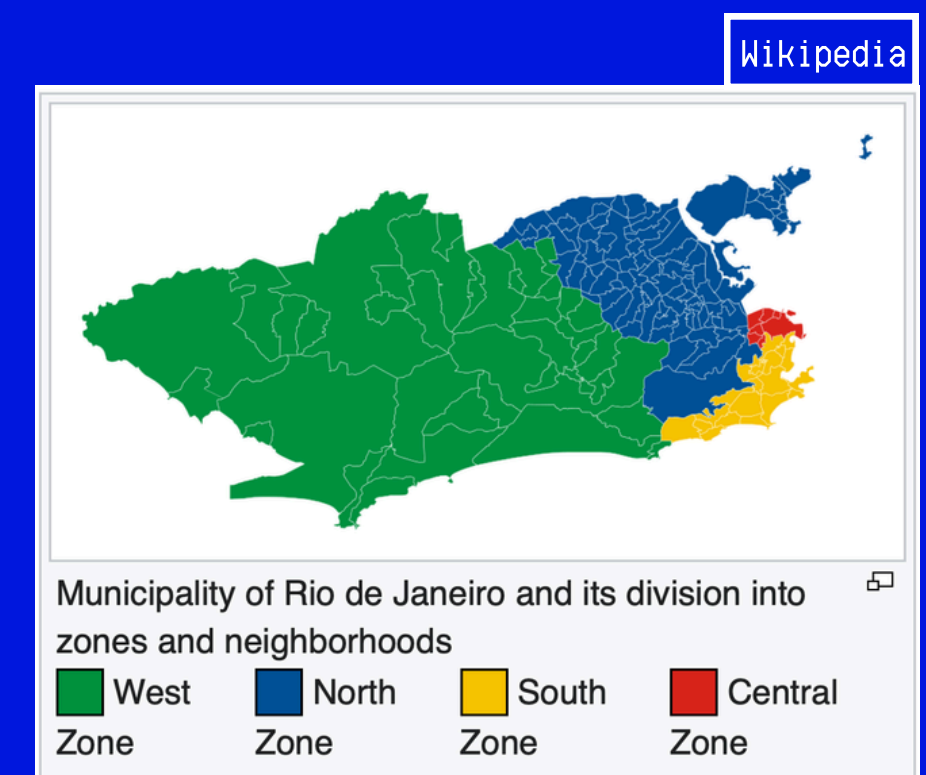
PROJECT OVERVIEW

INTRO

>>> Dataset: Inside Airbnb, Rio de Janeiro, June 24th 2026 snapshot;
>>> Business problem: helping a traveler choose where to stay, with data instead of guesswork;

HYPOTHESES

>>> South Zone has the most listings and best locations;
>>> Private rooms beat entire homes on value;
>>> Superhosts outperform;
>>> Demand peaks around Carnival and summer;



DATA ACQUISITION

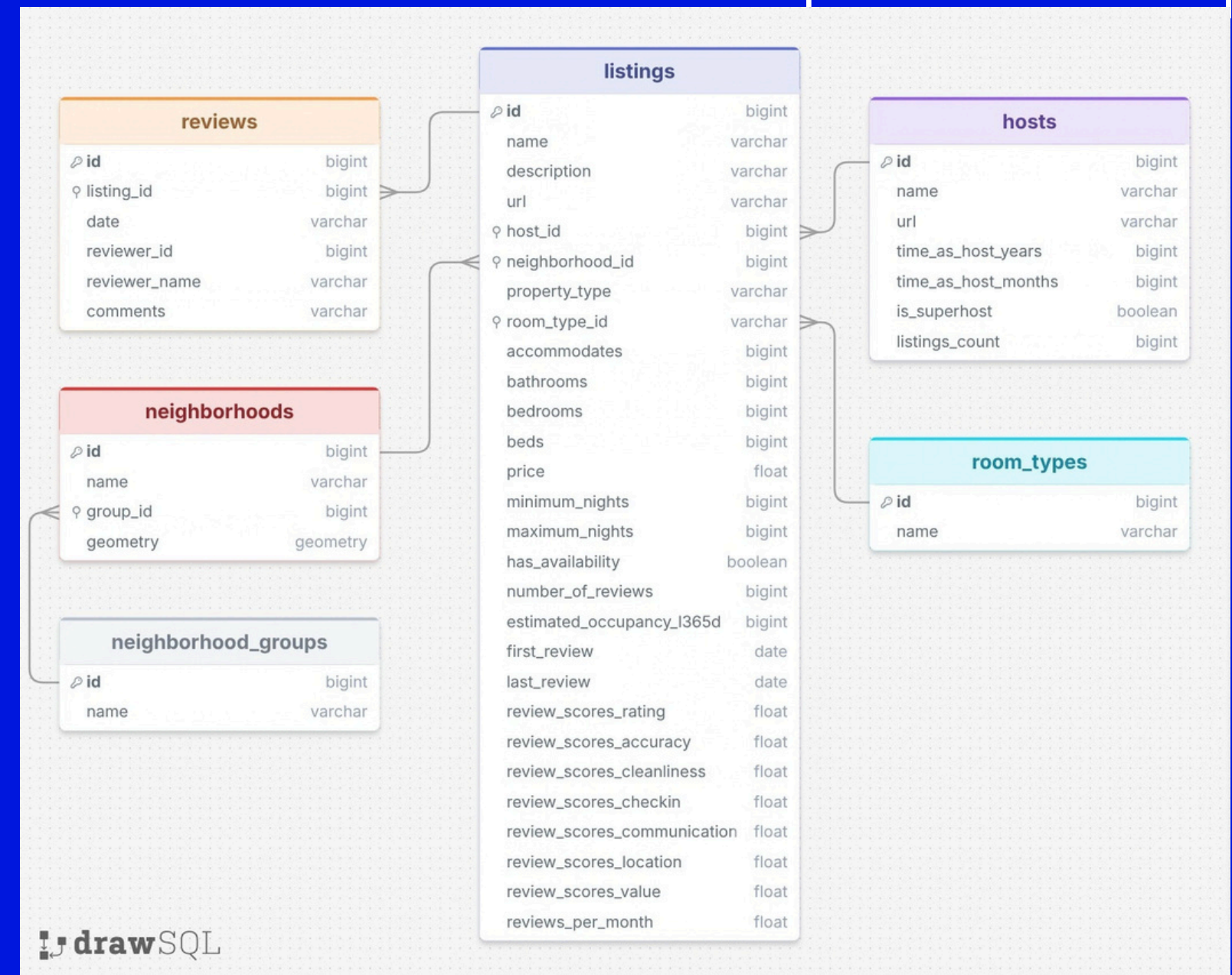
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>>> Data from Inside Airbnb,  
licensed under CC BY 4.0;  
>>> Challenge: no dataset linking  
Rio's neighborhoods to their zones;  
>>> Solution: converted a city-  
government PDF into a CSV
```

File	Rows	Columns	Description
listings.csv	48,713	90	One row per listing: price, room type, neighborhood, host, review scores, availability.
reviews.csv	1,388,472	6	One row per review: listing_id, date, reviewer_id, reviewer_name, comments.
neighbourhoods.csv	160	2	One row per neighborhood.
neighbourhoods.geojson	160	3	One row per neighborhood, with geometry.

DATABASE DESIGN

>>> 6 tables: listings (central),
 hosts, neighborhoods,
 neighborhood_groups, room_types,
 reviews
 >>> Normalized out repeated host,
 neighborhood, and room-type info
 into lookup tables

Entity-Relationship Diagram

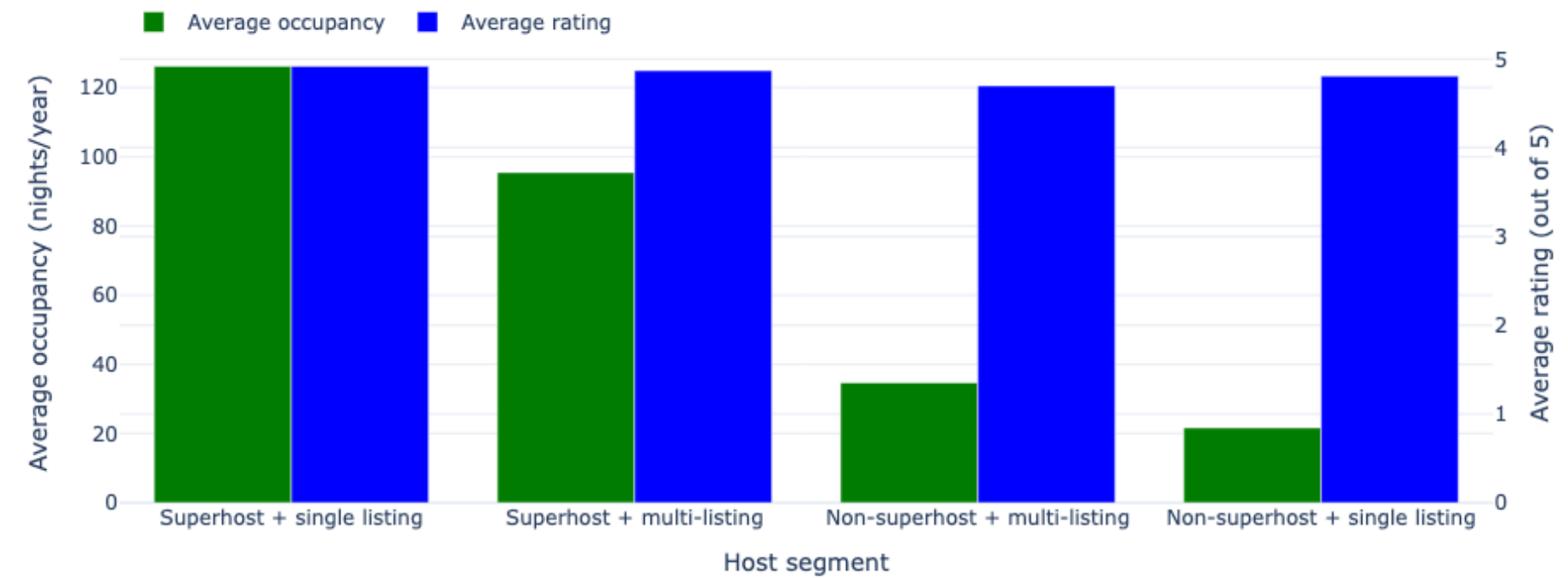


SQL INSIGHTS

FINDING

>>> Superhosts with a single listing outperform everyone – 126 nights of occupancy a year vs. 22-35 for others

Average Occupancy and Rating by Host Segment



VISUALIZATIONS

Let's see it on the
notebook.

CONCLUSIONS

>>> Confirmed: South Zone dominates listing count and location scores; Superhosts outperform

>>> Partially confirmed: value-for-money favors private rooms, but for larger groups, not smaller ones; Carnival peak only showed up clearly in 2025

>>> Surprise: Flamengo, not Copacabana, has the best location scores; November is the real seasonal peak

>>> Business implication: a smart traveler should look slightly off the most famous blocks, book a private room for a group, and travel outside the June-August window

OBSTACLES

>>> Challenge 1: foreign key mismatches when loading listings, hosts, and neighborhoods into SQLite

>>> Challenge 2: building a single combined ranking metric that fairly weighs price, rating, location and host reputation together

THANK YOU
FOR YOUR
ATTENTION!

ENDING 